



Fig. 2: Author's prototype

1 Hz or one pulse per second at its pin 13. The 1Hz pulse is fed to transistors T1 and T2. Transistor T3 is driven by a 256Hz audible tone from output Q6 of IC1. Transistors T2 and T3 are connected in series. As clock pulses are applied to the base of transistor T2, it conducts once every second. Transistor

T3 conducts to produce a tone from the speaker. LED1 also flashes for every clock pulse received.

Assemble the circuit on a general-purpose PCB and enclose in a cabinet. Mount the miniature speaker, LED1 and power switch S1 on the front panel of the cabinet. Switch S1 can be single-pole, single-throw type or a push-switch with push-on/push-off action. The circuit is powered by a 9V PP3 battery. It can also operate off a 6V or 12V battery.

This 1Hz clock timer can be used in counters, stop watches and other digital circuits. ●

Home Appliances Control Using Telephone

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Using this circuit, you can control up to seven electrical appliances through a telephone. The circuit can switch on/off the appliances in two modes. In the first mode, all seven or selected appliances can be switched on individually but switched off simultaneously by pressing a single button on the telephone's keypad. In the second mode, individual appliances can be switched on and off sequentially.

The circuit is built around dual-tone multiple-frequency (DTMF)

IC MT8870 (IC1), 8-bit addressable latch IC CD4099 (IC2), relay driver IC ULN2003 (IC3), voltage regulator IC 7805 (IC4), seven relays and a few discrete components. Connect the appliances to mains through the relays, e.g., bulb (load 1) to relay RL1, fan (load 2) to relay RL2 and so on. Also, assign digit keys to the appliances, say, key 1 to bulb, key 2 to fan, key 3 to television, etc.

Working of the circuit is simple. When you lift the handset from the cradle and press any key (1 through 7), the respective binary output of IC1 goes low while pin 15 goes high.



Transistor T1 conducts to enable the latch (IC2), making its corresponding output pin high. The binary outputs of IC1 (Q0 through Q2) are connected to the inputs of IC2 (A0 through A2). IC2 converts the binary input into its decimal equivalent, which is available at its output pins Q1 through Q7. The output of IC2 is fed to relay driver IC3.

Pressing of keys 1 to 7 (assigned to different loads) causes latching of the

